



Ionic self-diffusion of Na₂O–Al₂O₃–SiO₂ glasses from molecular dynamics simulations

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ARTICLE INFO

Keywords:

Aluminosilicate glass
MD simulation
Diffusion behavior
Effective rings

ABSTRACT

The structure of alkali aluminosilicate glasses (NAS) as a function of the molar ratio of Al/Na ($R_{Al/Na}$) were investigated by using classical molecular dynamics simulations and validated by comparing with the high energy X-ray diffraction experiments. General structural features, including the pair distribution functions, coordination number, Q_n distribution, oxygen species and ring size distributions, were analyzed to elucidate the effects of Al content on the structure of the glasses. By inducing effective rings size as a new parameter, the $R_{Al/Na}$ dependence of the alkali ion diffusion ability below the glass transition temperature were analyzed on the basis of medium range structures instead of the calculated diffusion coefficient. The results suggest that the optimum aluminum content for NAS system exists at around $R_{Al/Na} = 1.2$ not only with a high ionic self-diffusion but also with a good network connectivity, providing a theoretical guidance for the design and optimization of chemically strengthened glasses.

1. Introduction

The chemically strengthened glasses are widely used as protective display covers in commercial electronic devices due to their high strength and scratch resistance [1]. The ion exchange process for chemical strengthening was originally developed by Kistler [2] in 1962 and involved replacing smaller alkali ions such as Na⁺ from the glass by larger ions such as K⁺ from a surrounding molten salt bath at a temperature below the glass transition temperature T_g , inducing the formation of a compressive stress layer that tremendously improved the mechanical properties of the glass. Alkali aluminosilicate glasses are good candidates for producing chemically strengthened glasses as they undergo the least plastic deformation during the ion-exchange process and can develop a higher surface compression [3,4]. Local structural and topological environments around aluminum ions allow for high alkali mobility while maintaining a relatively rigid glass network [5]. As a result, with increasing Al₂O₃ content, surface compressive stress, surface hardness and K⁺/Na⁺ ion-exchange ratio increase during the chemically strengthened process as previously reported for peralkaline Na₂O–Al₂O₃–SiO₂ (NAS) compositions ($Al/Na < 1$) [6]. In silicate glasses, the added alumina mainly play the role of glass former and exist in the form of [AlO₄] tetrahedra, which link with [SiO₄]

tetrahedra through corner-sharing and enter the glass forming network. However, there is an important difference between [AlO₄] and [SiO₄] network forming units: each [AlO₄] tetrahedron contains one negative charge while [SiO₄] tetrahedron is electrically neutral. As a result, in aluminosilicate glasses, sodium ions can play two different roles: first as a network modifier by breaking up the glass network and second as a charge compensator around [AlO₄] tetrahedron to maintain the charge neutrality. The bond strength of Na⁺ with non-bridging oxygens from [SiO₄] or bridging oxygens from [AlO₄][−] is different and influences sodium ion mobility in the glass network.

Alkali ionic self-diffusion and interdiffusion behavior in NAS ternary system is of great importance to address the mechanism of chemical strengthening and further improve the mechanical properties of these glasses [3,7,8]. Although chemical strengthening technology has been known for decades, atomistic mechanisms ultimately responsible for ion exchange are not yet well-understood [4], especially for per-aluminous compositions ($Al/Na > 1$) [9–11].

Aluminosilicate glasses have been investigated by means of molecular dynamics (MD) simulation on local and medium length scales in conjunction with dynamic properties [12–17]. Ryan et al. [18] studied the thermodynamic and transport properties of NaAlSi₃O₈ glass melt and investigated ion diffusions at high pressure (30 GPa) and high

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<https://doi.org/10.1016/j.jnoncrysol.2019.119734>

Received 11 August 2019; Received in revised form 26 September 2019; Accepted 6 October 2019

Available online 31 October 2019

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temperature (from 3000 K to 5100 K) using MD simulations based on the mean square displacement (MSD) calculations. The results showed that the activation energies were respectively 85 and 140 kJ/mol for Na^+ and Si^{4+} , hence suggested MSD was an effective method to study dynamic properties of melts. However, chemical strengthening process often occurs at temperature around 700 K⁶, below the glass transition temperature (T_g), leading to difficulties of MSD calculations as an effective way to obtain diffusion coefficients.

Medium range structures such as primitive ring statistics play an important role on ionic diffusion. For example, in the study of helium diffusion in vitreous silica, Shen et al. [19] indicated that six- and larger-membered rings in glass structure provided the useful interstitial solubility sites, and the adsorption-induced deformation of vitreous silica could be explained by poroelastic theory [20]. These results are instructive to understand ion diffusion in silicate glasses. We might be able to treat NAS glasses as a porous and flexible media during chemical strengthening process. This may open a new way to understand ion transport properties in aluminosilicate glasses.

In the present study, we report the results of a systematic investigation of the effect of the molar ratio of Al/Na ($R_{\text{Al/Na}}$) on the structures and properties of glasses with composition (85-x) SiO_2 -x Al_2O_3 -15 Na_2O (mol%, $1 \leq x \leq 30$, corresponding to $0.067 \leq R_{\text{Al/Na}} \leq 2.000$). Detailed structural features in both short-range and intermediate-range, including nearest neighbor environments, oxygen speciation, and ring size distributions have been studied. Diffusion coefficients and activation energy barrier of sodium ions were calculated above glass transition temperature (T_g). By defining a new parameter, effective ring, the effect of the variation in the Al/Na ratio on the ion self-diffusion in both peralkaline and peraluminous compositions, has been discussed in details. Two more complex glass mixed alkali and alkali earth aluminosilicate glass compositions with different $R_{\text{Al/Na}}$ values were also studied for comparison.

2. Methodologies

2.1. Simulation details

Seventeen sodium aluminosilicate glass compositions with different $R_{\text{Al/Na}}$ were selected for MD simulation studies. The specific compositions are given in Table 1 and graphically shown in the ternary phase diagram in Fig. 1. The calculated densities [21–23] ranging from 2.35 to 2.58 g/cm³ are also shown in Table 1.

Two more complex aluminosilicate glass compositions with K_2O and alkali earth oxides, shown in Table 2, were also simulated to understand the effect of different $R_{\text{Al/Na}}$ on the glass structures. These two

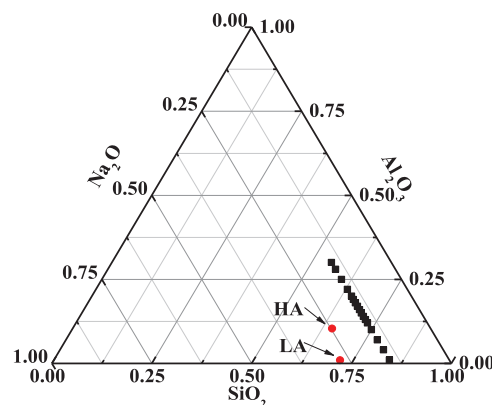


Fig. 1. Compositions of Na_2O - Al_2O_3 - SiO_2 system (black squares) and LA&HA samples (red circles) in ternary phase diagram. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Compositions (mol%) and densities of alkali and alkali earth aluminosilicate glasses with two different Al/Na ratios.

Glass	SiO_2	Al_2O_3	Na_2O	K_2O	CaO	MgO	Density(g/cm ³)
LA	71.66	0.94	12.75	0.49	8.83	5.33	2.50
HA	65.00	10.35	15.15	0.65	0.08	8.77	2.48

samples would be named as LA (lower alumina content) and HA (higher alumina content) in the following and were characterized experimentally that provided validation of the simulated glass structures.

A set of partial charge pairwise potentials were employed in our simulations with the charges of O, Si, Al, and Na assigned to -1.2, 2.4, 1.8 and 0.6. Assuming that the atoms were point charges with additional short-range forces acting on them, the rigid-ion model was used to describe solid based on the Born model. The short range interactions of the potential have the Buckingham form ($E_{ij} = A_{ij} \exp(-\frac{r}{\rho_{ij}}) - C_{ij}/r^6$, r is the distance between the two atoms and A , C and ρ are parameters), together with long-range Coulombic interactions. The parameters for this potential reported by Du and Cormack [24,25] were initially fitted to both structure and physical properties of various minerals. To avoid the negative infinity at small distances, a correction term is used to modify Buckingham potential due to much faster increase of the power term than the exponential term. This can be corrected by the addition of a spice function with the form of $V(r) = \frac{B}{r^n} + Dr$, where B , D , n are parameters. The spice function is applied for distance between atoms smaller than r_0 , where the second derivative of potential energy equals to 0 and the B , D and n parameters can be achieved when the potential, first derivative and second derivative of potential to be continuous at r_0 . Then Si-O, Na-O, and O-O pairs were modified.

The DL_POLY package [26] was employed for MD simulations. The computation of Coulombic interaction for each ion was completed by Ewald summation, using a cutoff of 8 Å to evaluate repulsive forces. In each simulation, around 10,000 atoms were used in the simulation box with applied periodic boundary condition so that chemical stoichiometry and density of each composition were satisfied as given in detail in Table 1. Time increment for the dynamics equations is 1 fs. All simulation runs were performed under the conditions of constant-volume-and-temperature (NVT) and microcanonical (NVE) ensembles with momentum and energy conserved within the tolerance of 10^{-6} . During the NVT ensemble, the Berendsen thermostat was used to control the temperature fluctuations. The simulations began with a randomly distributed configuration of atoms following energy minimized at 0 K to remove unreasonable structures, for example, too-closed atom pairs. Then the system was performed 100 ps NVT and 60 ps NVE runs

Table 1

Compositions (mol%) and densities of the simulated sodium aluminosilicate glasses.

Glass	SiO_2	Al_2O_3	Na_2O	Al/Na	Density (g/cm ³)
A1	84	1	15	0.067	2.35
A2	81	4	15	0.267	2.36
A3	78	7	15	0.467	2.38
A4	75	10	15	0.667	2.39
A5	73	12	15	0.800	2.40
A6	72	13	15	0.867	2.41
A7	71	14	15	0.930	2.41
A8	70	15	15	1.000	2.42
A9	69	16	15	1.067	2.43
A10	68	17	15	1.133	2.45
A11	67	18	15	1.200	2.46
A12	66	19	15	1.267	2.47
A13	65	20	15	1.333	2.48
A14	63	22	15	1.467	2.50
A15	60	25	15	1.667	2.53
A16	57	28	15	1.867	2.56
A17	55	30	15	2.000	2.58

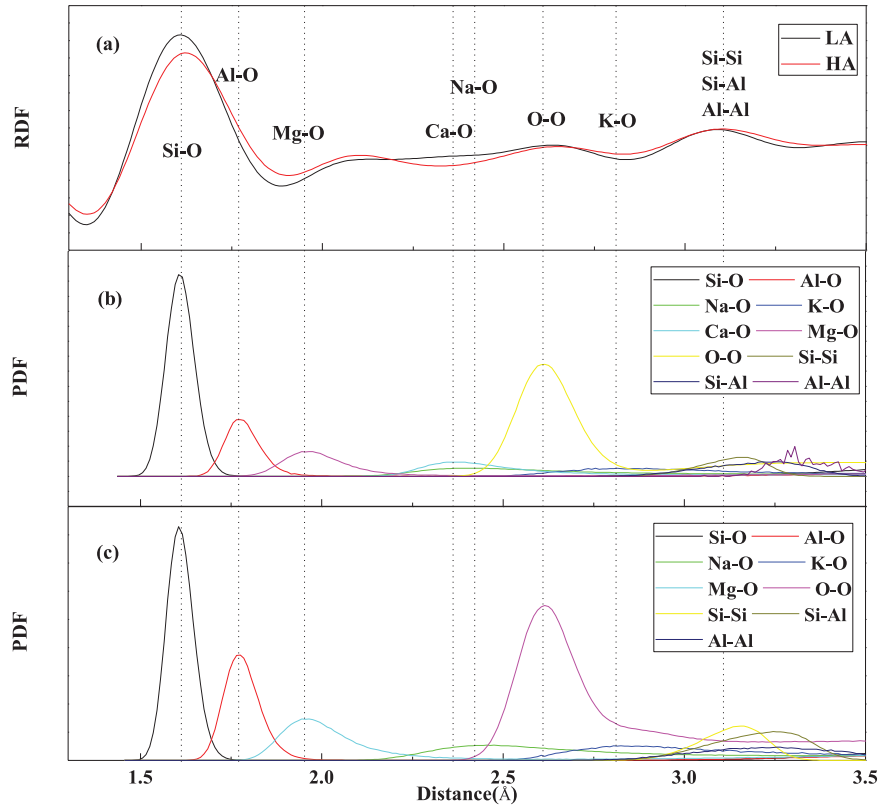


Fig. 2. Radial distribution function (RDF) from (a) high energy X-ray diffraction experiments and pair distribution function (PDF) of (b) LA and (c) HA compositions from molecular dynamic simulation.

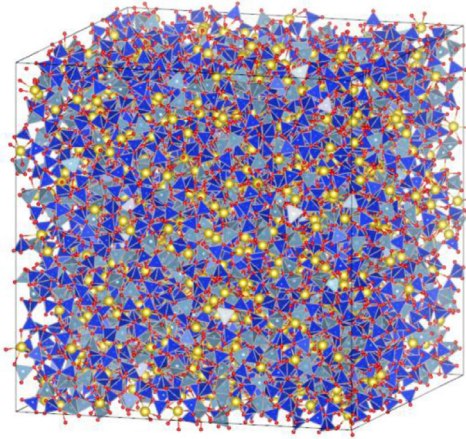


Fig. 3. Snapshot of the simulated glass structure when $R_{Al/Na} = 1$ (Blue tetrahedra: $[SiO_4]$; cyan tetrahedra: $[AlO_4]$; red balls: oxygen; yellow balls: sodium). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

at 6000 K to remove the memory effect of the initial configuration. After that, a stepwise cooling strategy [14' 17' 24' 25] was used: the system performed 100 ps NVT and 60 ps NVE runs to equilibrate at 4000, 3000, 2500, 2000, 1500, 1000, 750, 700, 650, 600, 550, 500, and 300 K. The data were collected every 200 fs during the last 40,000 steps at each stepwise cooling temperature for the following analysis. Note that three parallel simulations, for each composition, were conducted starting from different initial configurations to ensure the statistical reliability. The arithmetic mean value and standard deviation were used to present the result.

2.2. Experimental procedures

Two more complex aluminosilicate glasses, including K_2O and alkali earth oxides, with different contents of alumina (see in Table 2) were prepared. We performed high energy X-ray diffraction experiments at the beamline 11-ID-C of Advanced Photon Source (APS), Argonne National Laboratory (ANL) to determine nearest neighbor distances with a high degree of accuracy, compared with calculated data in Fig. 2. The general agreement between the simulations and experiments of these more complex aluminosilicate glasses indicates that the potential parameters work well for the $Na_2O-Al_2O_3-SiO_2$ system. Recent studies by comparing the structures of sodium silicate glasses from simulations using the same set of potential with solid state NMR further confirm the effectiveness of the potential to describe the structures of the sodium aluminosilicate glasses [27].

3. Results

3.1. General structural features of sodium aluminosilicate glasses

The structure analysis at room temperature was performed for aluminosilicate glasses with different Al/Na ratios from MD simulations. Fig. 3 shows a typical snapshot of the simulated glass structure ($R_{Al/Na} = 1$).

The short-range structures around Si and Al were well captured by the pair distribution function (PDF), bond angle distribution (BAD) and Q_n distribution. PDF, $g(r)$, is defined as $g(r) = \frac{1}{N\rho} \langle \sum_i^N \sum_{j \neq i}^N \delta(r - r_{ij}) \rangle$, in which N is the number of atoms, ρ is the number density, and δ is Dirac function. As r increases, $g(r)$ oscillates and approaches 1. The simulated PDFs of Si-O and Al-O are shown in Fig. 4. Coordination number is defined as $\int_0^{r_c} 4\pi r^2 \rho(r) dr$ in which the cutoff value r_c is 2.25 Å for both Si-O and Al-O. For all samples, silicon is 100% four

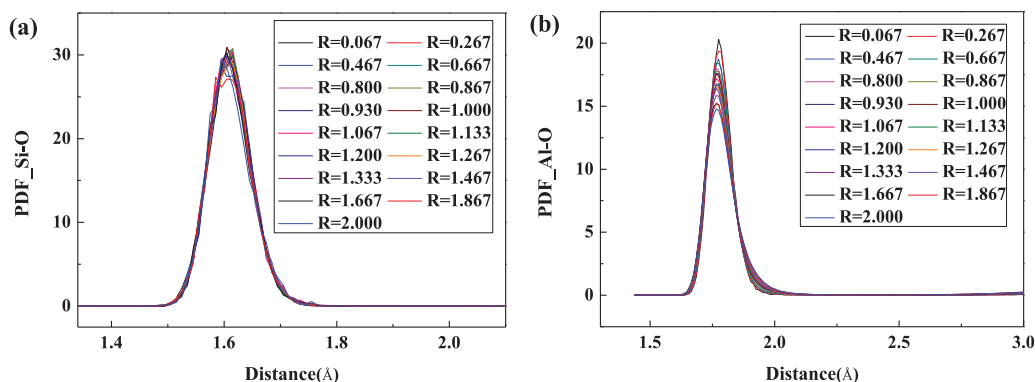


Fig. 4. Partial pair distribution functions of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass system (a) Si-O; (b) Al-O.

Table 3

The coordination distribution (%) of O around Al.

Glass	3	4	5	6
A1	0	99.558	0.442	0
A2	0.508	98.953	0.539	0
A3	0.164	99.334	0.502	0
A4	0.894	98.719	0.387	0
A5	1.045	98.551	0.403	0
A6	1.066	98.523	0.411	0
A7	1.144	98.452	0.404	0
A8	1.192	98.086	0.722	0
A9	1.213	98.291	0.496	0
A10	1.369	97.816	0.814	0.001
A11	1.181	98.386	0.432	0
A12	1.251	97.947	0.801	0
A13	0.778	98.706	0.514	0.002
A14	0.893	98.189	0.917	0.001
A15	0.888	97.785	1.326	0.001
A16	0.962	97.513	1.521	0.004
A17	0.793	97.499	1.704	0.004

coordinated to oxygen with an average Si-O bond length of 1.61 Å, suggesting that the local structure around Si is not sensitive to composition changes. Al-O PDFs have primary peaks located at around 1.76–1.77 Å. Compared to Si-O, the peak of Al-O is broader and with smaller amplitude, indicating weaker covalent feature of the Al-O bond. The coordination number analyses of oxygen around aluminum in Table 3 show that aluminum ions are mainly four coordinated in both peralkaline and peraluminous regions. The aluminum role in the aluminosilicate glass system relates to the macroscopic properties, such as density, viscosity, and compressibility [28]. It is generally accepted that the aluminum ions replace the silicon ions in $[\text{SiO}_4]$ tetrahedra and act as a network former in peralkaline compositions. According to our MD simulation results, the concentration of six-fold coordinated aluminum is very low, which is negligible within the statistical error, agreement with previous classic MD simulations [14]. We have observed the existence of five-fold coordinated aluminum in our simulation results, and the concentration increases from about 0.4% to about 1.7% with increasing Al/Na ratio, which is consistent with the recent high-resolution NMR experiment data with the concentration of five-fold coordinated aluminum changing from 0.5% to 2.0% [29]. It is worthy to note that as three-coordinated aluminum has not been detected with experimental methods, we regard the calculated $\text{Al}^{[3]}$ as the Al structural defects generated by the simulation method [14].

BAD analysis around silicon and aluminum is shown in Fig. 5. The O-Si-O bond angle, centered at about 108.2° (Fig. 5a), is not sensitive to the changes of Al content in glass compositions. The mean value is close to the perfect tetrahedron geometry at 109.5°, indicating the regularity of $[\text{SiO}_4]$ tetrahedron in NAS system. Compared to O-Si-O bond angles, the BAD of O-Al-O is broader and the intensity is lower, with a smaller mean value at about 107.2°, indicating the irregular

shape of $[\text{AlO}_4]$ tetrahedron (Fig. 5b). The inter-polyhedra bond angle distributions, including Si-O-Si, Si-O-Al and Al-O-Al, are also provided (see Fig. 5c-e). The intensity of all peaks varies monotonically with the addition of Al_2O_3 that is the amplitude of Si-O-Si decreases while that of Si-O-Al and Al-O-Al increases with the increase of Al/Na ratio. Like the bonding length of Si-O (Fig. 4a) and the shape of $[\text{SiO}_4]$ tetrahedra (Fig. 5a), BAD of O-Si-O is not sensitive to the Al content, indicating that interacting between $[\text{SiO}_4]$ changes little when the Al/Na ratio increases from 0.067 to 2.000. For BAD of Si-O-Al, with the changes in the Al/Na ratio from 0.067 to 2.000, the Si-O-Al decreased by approximately 10°. A decrease in the Al-O-Al bond angle of about 17° was observed that was the Al-O-Al main peak shifted from around 135° to around 118° when $R_{\text{Al/Na}}$ changed from 0.067 to 2.000. The edge-shared $[\text{AlO}_4]$ tetrahedra manifest themselves in the Al-O-Al bond angle distribution as the peak around 90° (see Fig. 5e), corresponding to the charge compensation in the peraluminous aluminosilicate glass compositions, consistent with the reported literature [14]. Although corner-sharing is the dominating method of connection, edge-sharing between $[\text{AlO}_4]$ tetrahedra does exist, and increases with increasing Al/Na ratio, accounting for the present and drastic increase of the peak around 90° for the BAD of Al-O-Al (Fig. 5e).

Q_n distribution of $[\text{SiO}_4]$ and $[\text{AlO}_4]$ tetrahedron is shown in Fig. 6. $[\text{SiO}_4]$ tetrahedra mainly exist as Q_4 species, with the coexistence of Q_2 and Q_3 groups. As shown in Fig. 6a, the amount of Q_4 for Si increased from around 69% to around 98% with the increase of $R_{\text{Al/Na}}$ ranging from 0.067 to 2.000, while the formation of Q_2 and Q_3 was inhibited, with Q_3 decreasing from around 28% to around 2% and Q_2 almost vanished when $R_{\text{Al/Na}} > 1$, consistent with the experimental results from the earlier literature [30]. As mentioned above, three coordinated aluminum was treated as the structural defects, therefore, we removed the influence of three-coordinated Al to obtain the corrected concentration of Q_3 speciation, as shown in Fig. 6b. For all Al/Na ratios, Q_4 of $[\text{AlO}_4]$ is larger than 96.75%. $[\text{AlO}_4]$ tetrahedra in Q_3 type share three oxygen atoms with the other tetrahedra while leaving the fourth oxygen connected to sodium as a NBO. Due to the excess sodium ions, a small amount of Q_3 in $[\text{AlO}_4]$ tetrahedra, less than 2.5%, exists in peralkaline compositions and decreases with the increasing of the Al/Na ratio, which is consistent with the ^{27}Al MAS NMR experiment results in the literature [10]. Network connectivity (NC), defined as $\text{NC} = \sum_{n=0}^4 x_n \cdot n$, where x_n is the percentage of the Q_n species ($n = 0, 1, 2, 3, 4, 5$) quantifies the level of connection of the glass forming network formers. The NC results of Si, Al and overall connectivity are shown in Fig. 7. With the increasing of Al_2O_3 , NC of silicon increases drastically and the growth rate becomes slow when $R_{\text{Al/Na}} > 1$, while the NC of aluminum has a slight decrease. As a result, the overall connectivity has a saturated value at around $R_{\text{Al/Na}} = 1.2$, indicating the best network connectivity.

Oxygen speciation plays an important role in NAS system. As shown in Fig. 8, there are three types of oxygen: non-bridging oxygen (NBO)

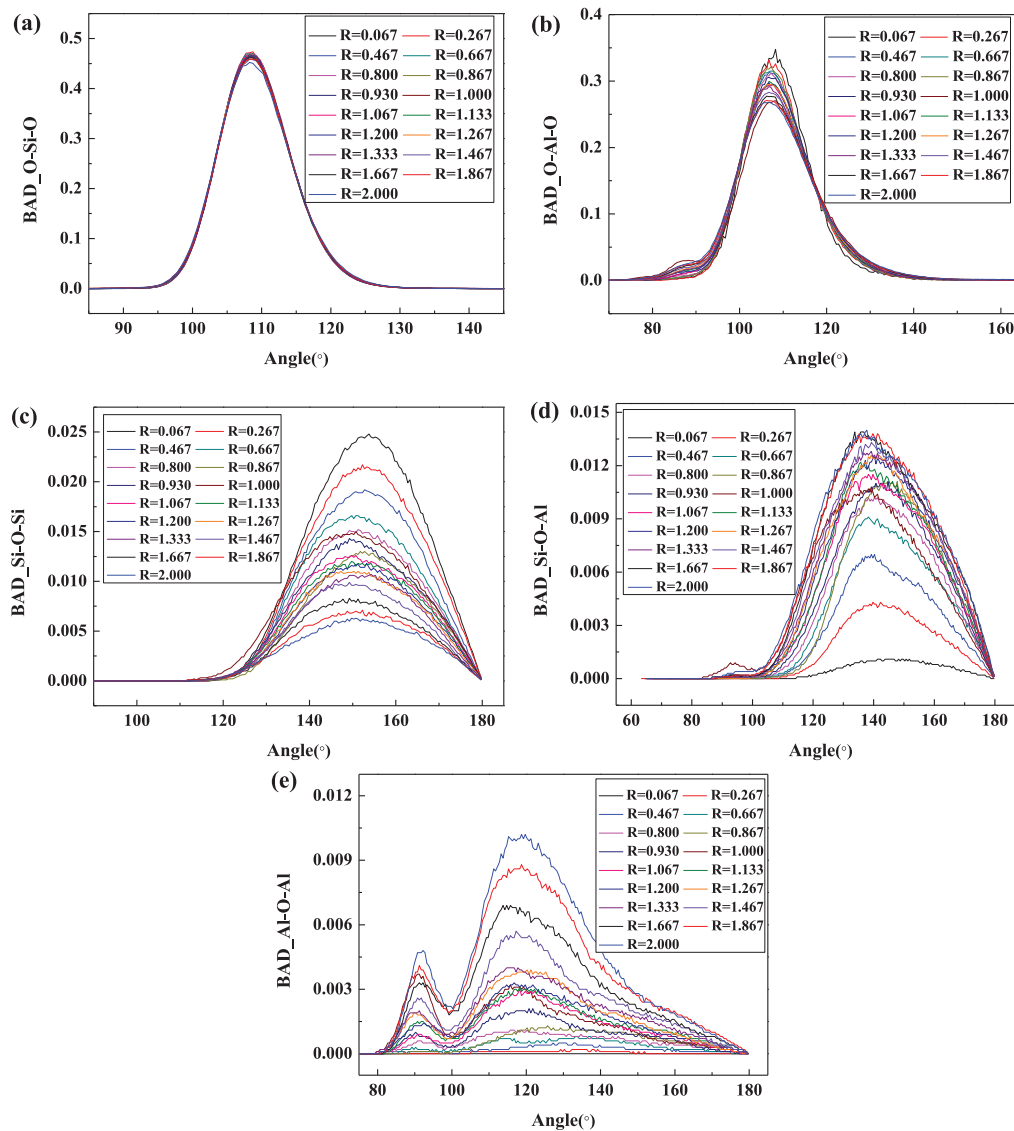


Fig. 5. Bond angle distributions of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass system (a) O-Si-O; (b) O-Al-O (c) Si-O-Si; (d) Si-O-Al; (e) Al-O-Al.

which connected to only one network former, bridging oxygen (BO) to two network formers, and three-bonded oxygen (TBO) to three network formers. NBOs have sodium ions associated with them. With the increasing Al/Na ratio, there is a monotonically decrease of NBO. Increasing the concentration of aluminum caused a transition between BO, NBO, and TBO. In peralkaline compositions, an increase in BO with the increasing Al/Na ratio, improving the connectivity of glass network. For the glass composition of $R_{\text{Al/Na}} = 1$, the concentration of NBO is less than 3% and BO is about 93%. After that, in peraluminous region, BO decreases with the increase of Al/Na ratio accompanied with the dramatic increase in TBO. In glass systems, oxygen triclusters have already been found in earlier NMR experiments [31] and supported by ab initio molecular orbital calculations [32]. For example, the presence of a high amount of oxygen triclusters in the amorphous $\text{Al}_2\text{O}_3-2\text{SiO}_2$ was found in which accounted for 30% of the total oxygen atoms [13]. Further quantity analyses of three types of TBO have been carried out as shown in Fig. 9. The type of 1Si2Al, which means TBO is bonded to one silicon and two aluminum ions (similar to 2Si1Al and 3Al), is dominating in compositions with $R_{\text{Al/Na}} < 1.5$. The fraction of 3Al type increases with the Al/Na ratio, while the coexistence of 2Si1Al is very few, showing a good agreement with the reported literature [14].

3.2. General structural features of alkali and alkali earth aluminosilicate glasses

More complex aluminosilicate glass compositions, LA and HA, were also simulated for comparison. Silicon and aluminum are almost 100% four coordinated to oxygen in both compositions with an average Si-O and Al-O bond length of 1.61 Å and 1.77 Å, respectively (see in Fig. 2b and c), consistent with the results from high energy X-ray diffraction experiment. The calculated Q_n of Si and Al in both LA and HA are shown in Fig. 10a, revealing that $[\text{SiO}_4]$ and $[\text{AlO}_4]$ tetrahedra mainly exist as Q_4 species. Furthermore, Q_4 of Si in HA is higher than that in LA while Q_2 and Q_3 show contrary behavior. Q_4 of Al has almost equivalent values in both LA and HA, indicating that is not sensitive to the Al/Na ratio, which agrees well with the results of NAS system. It is noteworthy that Q_3 of Al in HA is slightly higher than that in LA. Considering $R_{\text{Al/Na}}$ of HA is less than 0.5, such trend is consistent with the results as shown in Fig. 6b. Oxygen speciation has also been calculated in Fig. 10b. There are more bridging oxygens and less non-bridging oxygens in HA than those in LA, suggesting that the addition of Al_2O_3 increases the network connectivity in glass, consistent with the calculated NC in Fig. 7.

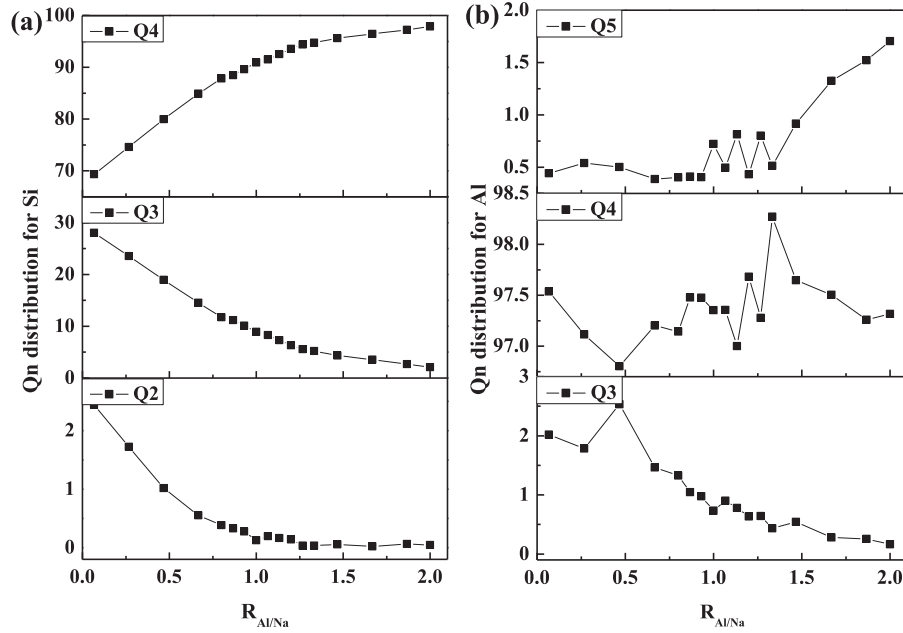


Fig. 6. Q_n of (a) Si and (b) Al in Na₂O–Al₂O₃–SiO₂ glass system.

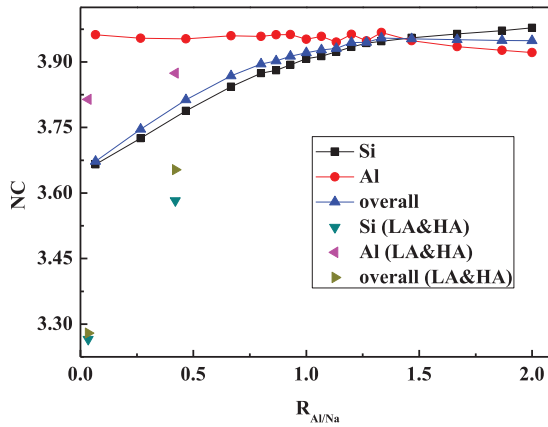


Fig. 7. Network connectivity of Na₂O–Al₂O₃–SiO₂ glass system and LA&HA compositions.

3.3. Diffusion behaviors of sodium ions in aluminosilicate glasses

Self-diffusion coefficient (D) of sodium ions was calculated from the mean square displacement (MSD) during the NVE simulation step via the Einstein equation:

$$D = \lim_{t \rightarrow \infty} \frac{\langle r^2(t) \rangle}{6t} \quad (1)$$

$$\langle r^2(t) \rangle = \frac{1}{n} < \sum_{i=1}^n |r_i(t) - r_i(t_0)|^2 \quad (2)$$

where $\langle r^2(t) \rangle$ is the MSD and t is the time.

Based on Arrhenius relationship of diffusion coefficient, we got:

$$\ln D = \ln D_0 - \frac{\Delta E_a}{RT} \quad (3)$$

where ΔE_a is the diffusion energy barrier, T is the temperature, and R is the gas constant.

The MSDs were calculated from NVE trajectories to quantify the sodium diffusion behavior in Na₂O–Al₂O₃–SiO₂ glass melts. All the sodium ions in each simulated structure were used and averaged over three parallel simulations. The logarithm scale of MSDs for sodium ions

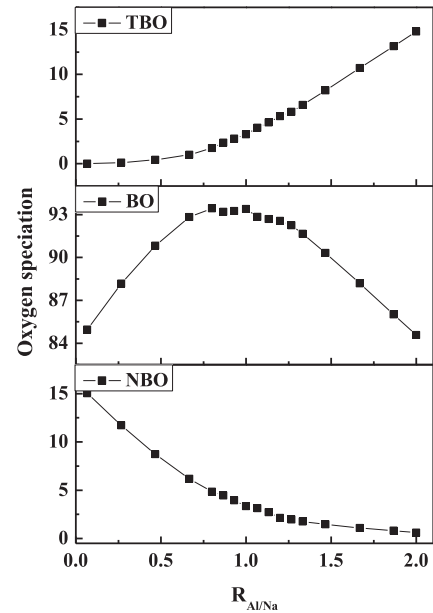


Fig. 8. Oxygen speciation in Na₂O–Al₂O₃–SiO₂ glass system.

is shown in Fig. 11. As expected, the relationship between MSD and time is strongly nonlinear at relative low temperatures ($T < 2000$ K) due to low diffusion coefficients, as shown in the inset of Fig. 11. This might induce uncertainties to the calculation of ionic self-diffusion at these low temperatures.

As a comparison, the sodium diffusion coefficients at high temperature (between 2000 K and 4000 K) were calculated as shown in Fig. 12. It can be seen that at higher temperatures when $T \geq 3000$ K, these samples are liquid and the melts have almost identical sodium diffusion coefficients. When it comes to lower temperatures ranging from 2000 K to 3000 K, there is a dispersion of diffusion coefficients with different R_{Al/Na}. Based on Eq. (3), the diffusion activation energy barriers, ΔE_a , were yielded as shown in Fig. 13. Compared with experimental result of 75.5 ± 1.2 kJ/mol⁸ and 85 kJ/mol¹⁸, the ΔE_a for composition with R_{Al/Na} = 1 from our simulation is 71.76 kJ/mol, in good agreement with experimental energy barriers.

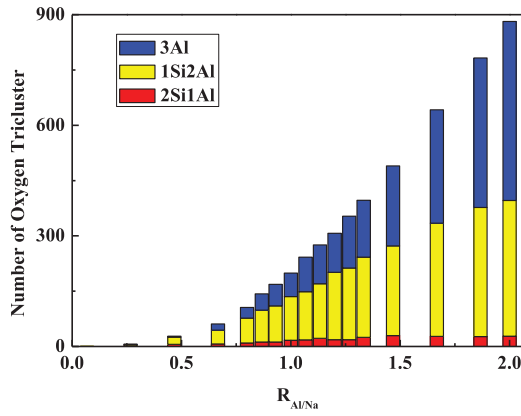


Fig. 9. Quantification of different types of oxygen triclusters.

3.4. Ring size distributions and medium range structures

Ring statistics is an important way to investigate the medium range order of glasses. Ring size distribution of a series of NAS glass is shown in Fig. 14a. As discussed before, the glass network is formed by connections of $[\text{SiO}_4]$ and $[\text{AlO}_4]$ units, therefore, the rings are composed of both Si and Al, with the size calculation based on the Si–O–Al connection and without treating Al–Al or Si–Si as part of the ring, that is, N-membered rings is named as the number of glass network formers contained in the ring. Using this ring analysis, we find that 5- to 8-membered rings are dominant for NAS system, while some larger membered-rings up to 12 members. With increasing Al/Na ratio, there are relatively more smaller-sized rings, i.e., 5 to 8 membered rings, and less larger-sized rings which contain 9 or more members, suggesting a tighter network structure. In peraluminous compositions, the amounts of tetrahedra and linkages between them keep increasing so that small-membered rings are formed at the cost of large rings. For example, for the composition with Al/Na ratio equal to 0.067, the largest ring contains 18 network forming atoms while the size of largest ring decreases to 12 members when the Al/Na ratio increases to 2. Non-bridging oxygen, accounting for the change of the size of largest ring in NAS system, can prevent the closure of rings and provide channels for alkali migration in glass network [10].

The ring size distribution of LA and HA compositions are shown in Fig. 14b. There are more 5- to 8-membered rings in HA than that in LA, showing a good agreement with the result of NAS system as shown in Fig. 14a. This trend can be explained by that more sodium ions in HA mainly play the role of charge compensator, causing the increase of bridging oxygen and improving the connectivity of glass network, accounting for the diffusion behavior of different compositions.

As for silica crystals, there are entirely 6-membered rings in cristobalite [33]. But silica glass contains mainly 4- to 8-membered rings

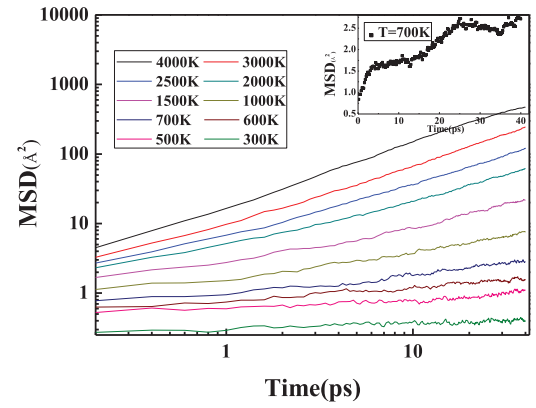


Fig. 11. Mean square displacements of sodium ions in glass composition of $R_{\text{Al/Na}} = 1$.

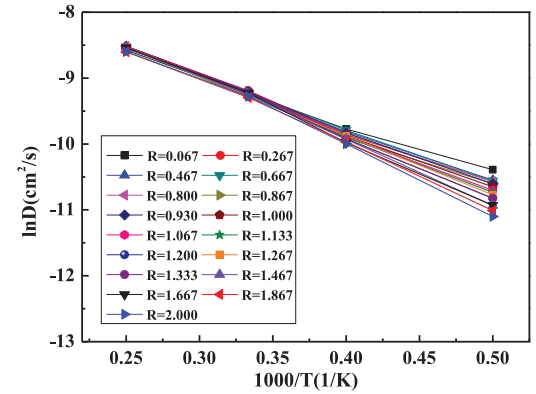


Fig. 12. Diffusion coefficients of sodium ions in $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass system from 2000 K to 4000 K.

with more disorder and open structures compared with crystals [34]. Assuming that the size of 6-membered rings in silicate glass is similar to that in β -cristobalite silica, the size of 6-membered rings in glass is estimated to be around 0.30 nm in diameter [19]. Considering that the kinetic diameters of sodium ion and potassium ion are 0.19 and 0.26 nm respectively, 6- and larger-membered rings can provide channels for Na^+/K^+ ion exchange. Therefore, we define effective rings for ion exchange process as 6- and larger-membered rings. Sum of all-sized rings and effective rings were calculated respectively in Fig. 15 with various $R_{\text{Al/Na}}$ for NAS glasses through the equations:

$$N_{\text{all-size}} = \sum_{i=1}^n R_i \quad (4)$$

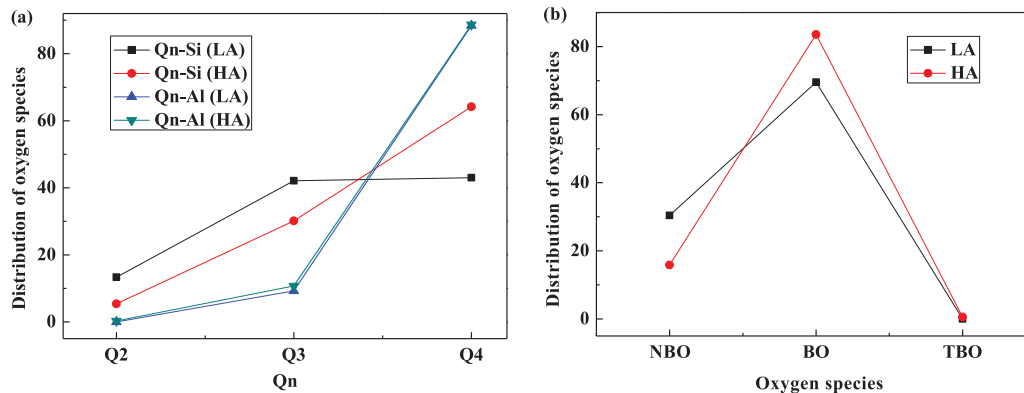


Fig. 10. Simulated structural information for LA and HA (a) Q_n distribution, (b) oxygen species.

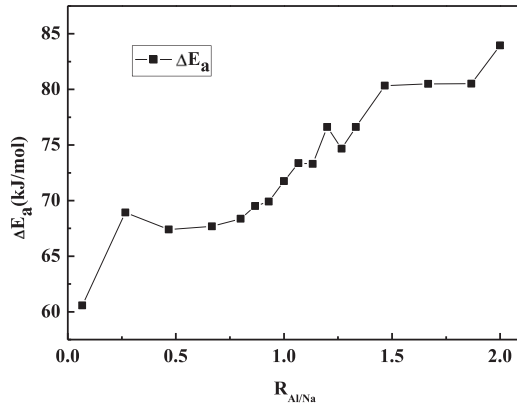


Fig. 13. Diffusion activation energy for $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass system.

$$N_{\text{effective}} = \sum_{i=6}^n R_i \quad (5)$$

where R_i is the number of i -membered rings per network former.

The total number of rings in glass network increase monotonically with the addition of Al content due to the decrease of NBO and the increase of BO. The value of effective rings first increases with $R_{\text{Al/Na}}$ and then reaches a plateau in peraluminous compositions with $R_{\text{Al/Na}}$ of around 1.2.

4. Discussion

In this work, comparison of pair distribution function between high energy X-ray diffraction experiments and molecular dynamics simulation showed general agreement of two complex aluminosilicate glasses LA and HA, demonstrating that the potential parameters work well for the $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ system. Short range structures, including PDF, coordination number and BAD, indicate that the addition of alumina mainly affects the interaction between inter-tetrahedra and not the individual tetrahedra. Thus medium range structure analyses apparently depend on the composition change. Q_n distribution for aluminum demonstrates that most of the $[\text{AlO}_4]$ tetrahedra connect themselves to other four tetrahedra and prefer to exist in Q_4 type of connection with neighboring glass forming units. With the addition of Al_2O_3 , more sodium ions play the role of charge compensator and less to break the linkage between the network formers, consulting the increasing amount of bridging oxygen and network connectivity with the increasing Al/Na ratio in peralkaline compositions. In peraluminous region, the increasing TBO supports the earlier proposed mechanism for neutralization of the extra negative charge around $[\text{AlO}_4]$ when there is not enough Na^+ for charge compensation. Detailed study of TBO in aluminosilicate glass has been performed and found that most of them

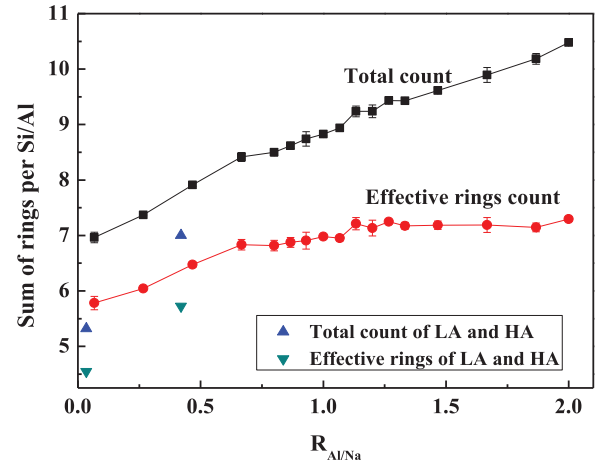


Fig. 15. Sum of rings per Si/Al in $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass system as a function of molar ratio of Al/Na.

are involved in two-membered or three-membered rings [13]. It is worth to note that in peraluminous region NBO still exists, agreement with the previous reported results in which a maximum value of NAS glass viscosity occurred in the peraluminous region rather than at the critical point of the Al/Na ratio equal to 1 [35]. Structure analysis of complex compositions LA and HA with different Al content have further shown that aluminum ions mainly form tetrahedra with oxygen but the packing of $[\text{AlO}_4]$ units is different from that of $[\text{SiO}_4]$ units, showing the same trend with the designed NAS glass compositions discussed above.

The results of MSD calculated from 300 K to 4000 K indicated that was an effective method to study dynamic properties of melts with high temperature but not applicable to the situation below the glass transition temperature when chemical strengthening process often occurred [6]. Based on the ring size distribution, effective rings for ion exchange have been induced. When the Al/Na ratio is larger than 1, the amount of TBOs increases, together with those of two- or three-membered rings. In addition, high Al content causes an increase of the amount of edge-sharing tetrahedra reflected by the peak at around 90° in the Al–O–Al bond angle distribution, accompanying the generation of 2-membered rings. However, these small rings cannot provide channels for ion exchange. As the effective rings reach the plateau value at around $R_{\text{Al/Na}} = 1.2$, glass composition at the R value might present a compromise between high network connectivity with large enough rings for sodium ion diffusion and over tightening of the network that creates TBO and small membered rings thus provide the optimal composition for ion diffusion and ion exchange.

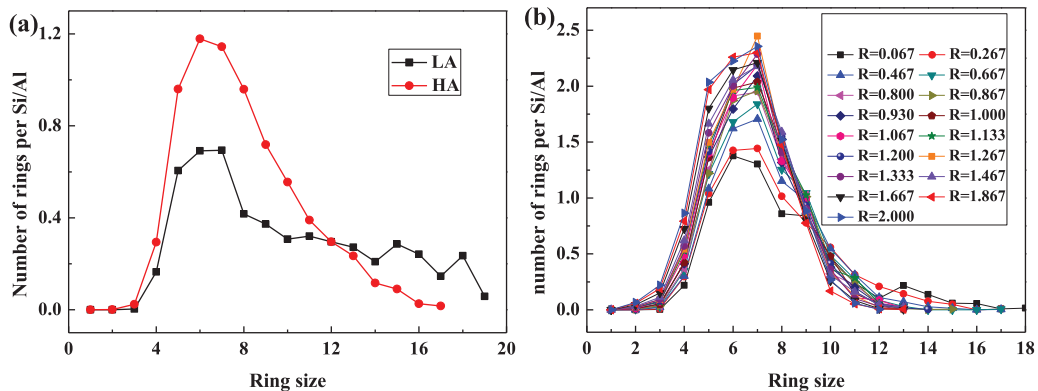


Fig. 14. Ring size distribution of (a) $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glass system, (b) LA and HA compositions.

5. Conclusions

We have investigated the structure and ion transport properties of a series of $\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$ glasses as a function of different Al/Na ratios using classical molecular dynamics simulations. Two more complex glasses with mixed alkali and mixed alkali earth oxide were also studied with both MD and high energy X-ray diffraction. Good agreement was observed with experimental structure factors and distribution functions, suggesting validity of the potential models and simulation procedure employed. The simulated results show that $[\text{AlO}_4]$ is the dominant aluminum species, and higher coordinated alumina is negligible. The addition of alumina would improve glass network connectivity and an optimum value exists at around $R_{\text{Al/Na}} = 1.2$. As for aluminosilicate glass, 6- and larger-membered rings are defined as effective rings for ion exchange process. The amount of effective rings monotonically increases with the addition of alumina in peralkaline compositions, accompanying with the improvement of network connectivity. However, when it comes to peraluminous region, effective rings reach a saturated value due to the increase of oxygen triclusters and edge-sharing tetrahedra which cause the generation of small-membered rings (2- and 3-membered rings), correlative to the result of network connectivity. Therefore, in the present simulations, the optimum value of Al content, both for network connectivity and Na^+/K^+ ions diffusion, is demonstrated to be around $R_{\text{Al/Na}} = 1.2$. These results may provide theoretical guidance for the design and optimization of glass compositions for chemically strengthening.

Declaration of Competing Interest

There is no conflict of interest with this work.

Acknowledgment

This work was supported by the 13th Five-Year National Key R&D Program (No. 2016YFB0303700), National Natural Science Foundation of China (U1809217), Opening Project of State Key Laboratory of Advanced Technology for Float Glass and the Fundamental Research Funds for the Central Universities. We would like to thank Prof. Qiaoshi Zeng for assistance in using beamline APS and analyzing the experimental data.

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